Homework Assignment 3: Critical Systems by Diffusion, P_N and S_N

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Question 1: Critical radius of a reflected sphere

Question 1 is the one *diffusion* question of this assignment — the S_N machinery set out before Question 3 belongs to Questions 3–5 — and the only one with two materials. A multiplying core of radius R_C is wrapped in $d \in \{1, 2, 3, 10\}$ mean free paths of reflector, and the cross sections are again the Question 5 table of Assignment 1.

The two regions

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) = \frac{1}{r}\frac{d^2u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0}\,u = 0 \tag{1}$$

The core has $c_C > 1$, so the coefficient is negative and the solution oscillates; finiteness at the origin selects the sine. The reflector has $c_R \le 1$ and decays. With

$$B = k_0(c_C)\,\Sigma_{t,C}, \qquad \kappa = \frac{\Sigma_{t,R}}{\nu_0(c_R)}, \qquad R_{\text{ext}} = R_R + \frac{z_0(c_R)}{\Sigma_{t,R}} \tag{2}$$

the two profiles are

$$\phi_C(r) = A\,\frac{\sin(Br)}{r}, \qquad \phi_R(r) = F\,\frac{\sinh[\kappa\,(R_{\text{ext}} - r)]}{r} \tag{3}$$

k_0 and ν_0 are Case's discrete eigenvalue on its two branches, the same roots Assignment 1 used: $c\arctan(k_0) = k_0$ above $c = 1$ and $\operatorname{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ below it.

Matching, and the one criticality equation

The interface at $r = R_C$ carries a flux relation $\phi_R = g\,\phi_C$ and continuity of the net current $J = -D\,d\phi/dr$. Since $\phi = u/r$ gives $\phi'/\phi = u'/u - 1/r$, and the two logarithmic derivatives are $u_C'/u_C = B\cot(BR_C)$ and $u_R'/u_R = -\kappa\coth(\kappa L)$ with $L = R_{\text{ext}} - R_C$, all three approximations reduce to a single transcendental equation for R_C:

$$\boxed{D_C B\cot(BR_C) \;+\; g\,D_R\kappa\coth(\kappa L) \;-\; \frac{D_C - g\,D_R}{R_C} \;=\; 0} \tag{4}$$

with $\kappa L = [d + z_0(c_R)]/\nu_0(c_R)$, so the reflector enters only through its *optical* thickness. The last term has no slab counterpart: ϕ falls as $1/r$ even where u is flat, so a jump in D at a curved interface drives a current by itself. Equation (4) runs from $+\infty$ at $R_C \to 0$ to $-\infty$ at $R_C = \pi/B$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/B)$ and `brentq` finds it in one call.

The three approximations differ only in what they put into D_0, z_0 and g:

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	$1/3$	$\lvert 1-c\rvert\,\nu_0^2$	$\lvert 1-c\rvert\,\nu_0^2$
relaxation rate	$\sqrt{3\lvert 1-c\rvert}$	$k_0(c)$ or $1/\nu_0(c)$	$k_0(c)$ or $1/\nu_0(c)$
$z_0(c)$	$2/3$	$0.710446\,[1+q(1-c)^2]/c$	$0.710446\,[1+q(1-c)^2]/c$
$g = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: Only the last row makes (c) different from (b). Since $\mu_0 = 1/2$ for every medium in the classic theory, (a) is automatically continuous, and with no reflector at all (b) and (c) are the same bare sphere.

Zimmerman's μ_0. The discontinuity comes from Zimmerman (UCRL-82792), who writes the partial currents of the asymptotic mode as $\phi_\pm = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{c\,\nu_0^2}{2}\ln\!\left(\frac{\nu_0^2}{\nu_0^2-1}\right) \xrightarrow[c>1]{} \frac{c}{2k_0^2}\ln\!\left(1+k_0^2\right) \tag{5}$$

the second form being the exact analytic continuation through $\nu_0 = i/k_0$. It is not an ansatz: integrating the discrete mode $\psi(\mu) \propto (\nu_0-\mu)^{-1}$ over the two half-ranges reproduces it identically, and it returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam. Matching the traffic rather than the flux, $\mu_0(c_C)\phi_C = \mu_0(c_R)\phi_R$, is what makes ϕ itself jump. Nothing is lost at the interface — the same current crosses both faces.

material	c	mfp [cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H_2O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials. Sodium is a pure scatterer, $\Sigma_c = \Sigma_f = 0$, so $c_R = 1$ exactly: ν_0 diverges, $\kappa = 0$, and the reflector solution degenerates to u_R linear in r with $\kappa\coth(\kappa L) \to 1/L$. The code takes that limit explicitly rather than dividing out a near-zero κ.

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.5400	9.879	7.968	6.154
Water	2	5.0855	4.7987	4.3823	8.649	7.267	5.535
Water	3	5.0153	4.7499	4.3321	8.296	7.048	5.347
Water	10	4.9814	4.7244	4.3061	8.129	6.935	5.251
Iron	1	5.4212	4.9575	4.6450	10.478	8.013	6.591
Iron	2	5.1056	4.7364	4.4182	8.753	6.988	5.672
Iron	3	4.9506	4.6237	4.3059	7.979	6.501	5.250
Iron	10	4.6790	4.4188	4.1070	6.737	5.674	4.556
Sodium	1	6.4397	5.7159	5.5171	17.562	12.281	11.044
Sodium	2	6.3134	5.6405	5.4318	16.549	11.801	10.540
Sodium	3	6.2505	5.6022	5.3892	16.059	11.563	10.293
Sodium	10	6.1315	5.5286	5.3080	15.159	11.113	9.835

Table 4: Pu-239 core: critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion. Bare: 5.8163 cm / 12.940 kg classic, 5.1890 cm / 9.188 kg asymptotic.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.5228	30.330	26.273	22.087
Water	2	6.9185	6.6731	6.2694	26.355	23.649	19.611
Water	3	6.8146	6.5937	6.1865	25.186	22.816	18.844
Water	10	6.7638	6.5520	6.1432	24.627	22.385	18.451
Iron	1	7.2811	6.8392	6.5656	30.720	25.460	22.525
Iron	2	6.8002	6.4625	6.1789	25.027	21.480	18.775
Iron	3	6.5545	6.2644	5.9804	22.411	19.565	17.023
Iron	10	6.1125	5.8970	5.6199	18.176	16.320	14.126
Sodium	1	8.4264	7.7951	7.6194	47.618	37.697	35.205
Sodium	2	8.2322	7.6564	7.4692	44.401	35.721	33.164
Sodium	3	8.1328	7.5844	7.3919	42.811	34.722	32.145
Sodium	10	7.9401	7.4425	7.2414	39.840	32.809	30.221

Table 5: U-235 core, same columns. Bare: 8.1031 cm / 42.345 kg classic, 7.4339 cm / 32.696 kg asymptotic.

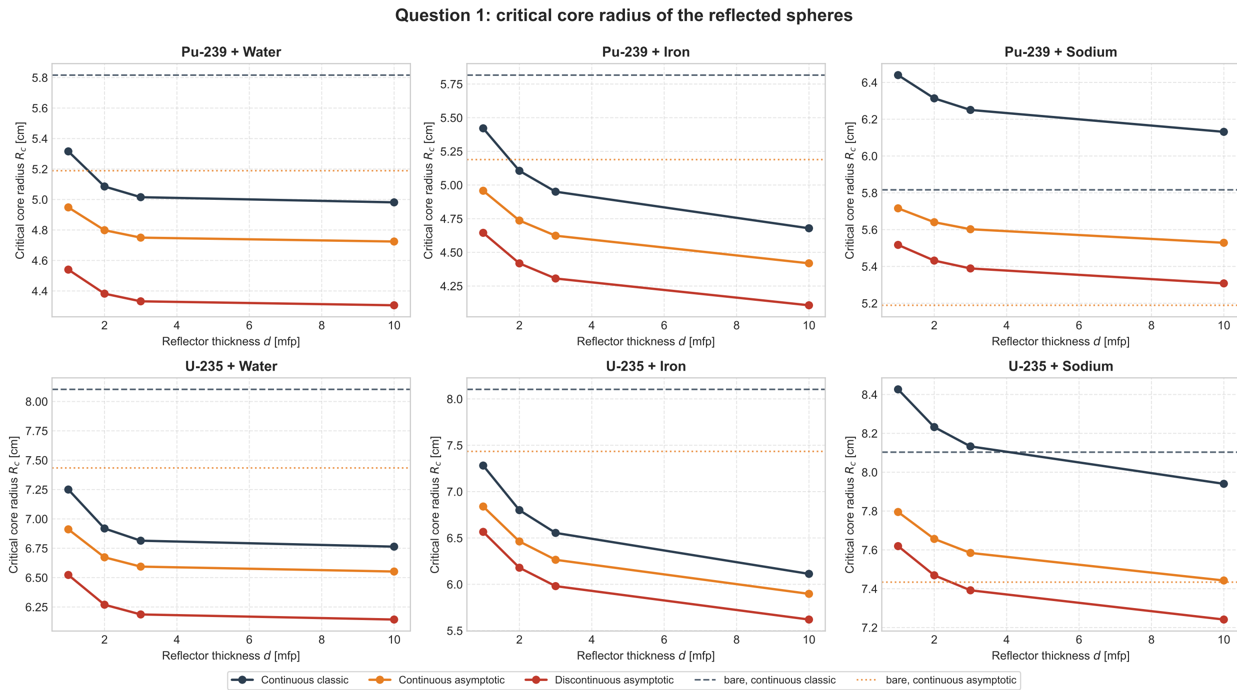

Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d \approx 3$ mfp; iron is still improving at $d = 10$; sodium sits above its bare line throughout.

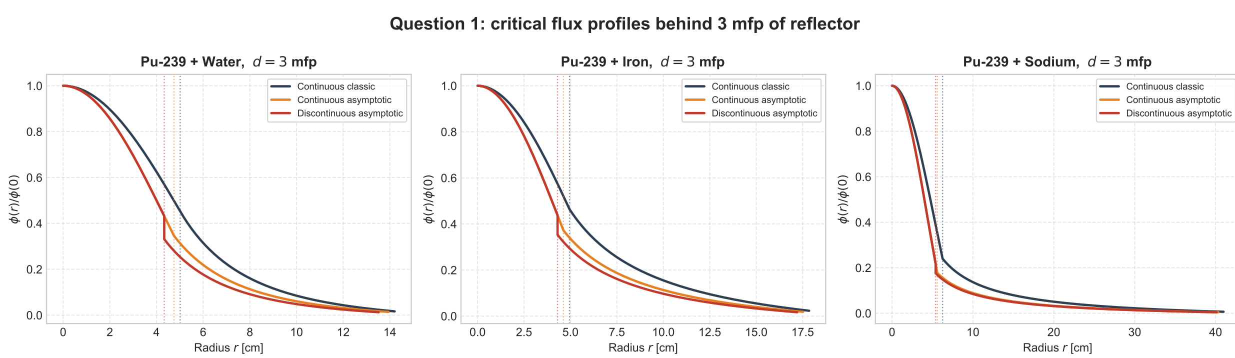

Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$; dotted verticals mark each theory's interface. The discontinuous curve drops by the factor g at the interface and is continuous in current, not in flux. Note the horizontal scale of the sodium panel: 3 mfp of sodium is 34.7 cm.

What the three approximations say

The reflector saving. Water and iron behave as reflectors should. Both cores lose 12–17% of their radius behind water and 12–24% behind iron by theory (c), which because the mass goes as R^3 is a factor of nearly two in critical mass: Pu-239 falls from 9.188 kg bare to 4.556 kg behind 10 mfp of iron. The two reflectors saturate differently, and the reason is c_R. Water absorbs ($c_R = 0.90$, $\nu_0 = 1.90$), so a neutron more than about two mean free paths deep never returns and the water columns are flat past $d = 3$. Iron barely absorbs ($c_R = 0.998$, $\nu_0 = 12.9$), so its return keeps improving out to $d = 10$ and beyond — iron starts as the worse reflector at $d = 1$ and ends the better one at $d = 10$.

The ordering of the three theories. At every entry $R_c^{(a)} > R_c^{(b)} > R_c^{(c)}$. The first gap is inherited from the bare problem: the asymptotic $D_0(c_C)$ and $k_0(c_C)$ already give a smaller sphere than $D_0 = 1/3$ does, by 11% for Pu-239. The second is the jump itself. Every pair here has $\mu_0(c_C) < \mu_0(c_R)$, so $g < 1$ — 0.766 for Pu-239 behind water, 0.873 for U-235 behind iron — because the multiplying core has the shorter relaxation length and hence the more forward-peaked angular flux, and so carries more current per unit density. A g below one multiplies down both the leakage term and the curvature term of (4), which is why (c) always predicts the smallest core, by a further 8–9% for water and iron.

The sodium columns are diffusion theory failing. Every sodium radius above is *larger* than the corresponding bare sphere, which no reflector can do. The condition is being solved correctly — (4) was checked against an independent two-region spherical finite-volume k-eigenvalue solve, 3000 cells with harmonic-mean face diffusion coefficients across the material jump, and the two agree to better than 0.4% on exactly these cases. The fault is in the model. Splitting the interface leakage of the Pu-239 core at $d = 10$ mfp into its two parts, $J/\phi = D_R/R_c + D_R/L$:

reflector	D_R [cm]	shell [cm]	geometric D_R/R_c	far-face D_R/L	total vs. bare 0.5592
Water	1.021	30.6	0.2050	0.1826	0.3876
Iron	1.433	43.0	0.3063	0.0381	0.3444
Sodium	3.859	115.8	**0.6295**	0.0312	0.6607

Table 6: Leakage ratio at the interface, in cm^{-1}, classic theory. Sodium's far-face return is the best of the three — 10 mfp of a pure scatterer really does send almost everything back — but its geometric term alone exceeds what a bare vacuum surface leaks.

The geometric term is the current driven by the $1/r$ spreading of $\phi = u/r$, and it is harmless only while $D_R \ll R_c$. Sodium's $\Sigma_t = 0.086368\ \mathrm{cm}^{-1}$ is a mean free path of 11.58 cm against a core radius near 6 cm: a neutron entering the reflector travels two core radii before its first collision, which is precisely the regime Fick's law expands away from. None of the three approximations repairs it, because a pure scatterer has $D_0 = 1/3$ in all of them and $D_R = 3.86$ cm throughout. Measured against its own bare sphere at $d = 10$, (a) is +5.4%, (b) +6.6% and (c) +2.3% — only the Zimmerman jump helps, and only because $g = 0.807$ multiplies the offending term. Transport theory would not make this mistake: the shell is optically thick and purely scattering, so its true albedo is high and the true critical radius must lie below the bare 5.1890 cm.

Question 2: Critical slab half-thickness by P_N

A homogeneous slab of total thickness a, reflective at the midplane $x = 0$ and vacuum at $x = a/2$, in mean free paths ($\Sigma_t = 1$), for $c \in \{1.2, 1.5, 1.8\}$ and $N \in \{1, 3, 5, 9\}$. The unknown is the critical half-thickness $a/2$ at which $k_{\text{eff}} = 1$ — the same quantity Question 3 tabulates for S_N, written the same way, so the two tables compare entry by entry.

The P_N equations

With isotropic scattering and fission lumped as in the paragraph "The meaning of c" before Question 3, and the fission source divided by k,

$$\mu \frac{\partial \psi}{\partial x} + \psi(x, \mu) = \frac{c}{2k}\, \phi_0(x), \qquad \phi_n(x) = \int_{-1}^{1} P_n(\mu)\, \psi(x, \mu)\, d\mu \tag{6}$$

Expanding $\psi(x, \mu) = \sum_{n \geq 0} \frac{2n+1}{2} \phi_n(x) P_n(\mu)$, multiplying (6) by $P_n(\mu)$ and integrating over μ turns the streaming term into a three-term coupling through the Legendre recurrence $(2n+1)\mu P_n = (n+1)P_{n+1} + nP_{n-1}$, while the collision term is diagonal and the isotropic source reaches only $n = 0$:

$$\frac{n+1}{2n+1} \frac{d\phi_{n+1}}{dx} + \frac{n}{2n+1} \frac{d\phi_{n-1}}{dx} + \phi_n(x) = \delta_{n0}\, \frac{c}{k}\, \phi_0(x), \qquad n = 0, \ldots, N \tag{7}$$

The system closes on itself once $\phi_{N+1} \equiv 0$ is imposed, which is the entire content of the P_N approximation. In vector form, with $\mathbf{\Phi} = (\phi_0, \ldots, \phi_N)^T$,

$$\tilde{\mathbf{A}}\, \frac{d\mathbf{\Phi}}{dx} + \mathbf{\Sigma}(k)\, \mathbf{\Phi} = 0, \qquad \tilde{A}_{n,n+1} = \frac{n+1}{2n+1}, \quad \tilde{A}_{n,n-1} = \frac{n}{2n+1}, \qquad \mathbf{\Sigma}(k) = \mathbf{I} - \frac{c}{k}\, \mathbf{e}_0 \mathbf{e}_0^T \tag{8}$$

$\tilde{\mathbf{A}}$ is the Jacobi matrix of the Legendre recurrence truncated at order N, so its $N + 1$ eigenvalues are exactly the roots of $P_{N+1}(\mu)$ — the Gauss–Legendre ordinates of S_{N+1}. That is the algebraic content of the claim in Question 3 that the two methods share their interior equations and differ only at the boundary.

Boundary conditions

At the midplane the flux is even in μ, so every odd moment vanishes:

$$\phi_{2m+1}(0) = 0, \qquad m = 0, \ldots, \tfrac{N-1}{2} \tag{9}$$

Equivalently $d\phi_{2m}/dx|_0 = 0$, which (7) then delivers for free; only one of the two families may be imposed or the problem is over-determined.

At $x = a/2$ no neutron enters, $\psi(a/2, \mu) = 0$ for $\mu < 0$, which $N + 1$ moments cannot satisfy pointwise. Marshak's choice is to annihilate the $\frac{N+1}{2}$ lowest odd moments of the incoming half-range,

$$\int_{-1}^{0} \mu^{2m-1} \psi(\tfrac{a}{2}, \mu)\, d\mu = 0 \implies \sum_{n=0}^{N} c_{mn}\, \phi_n(\tfrac{a}{2}) = 0, \qquad c_{mn} = \frac{2n+1}{2} \int_{-1}^{0} \mu^{2m-1} P_n(\mu)\, d\mu \tag{10}$$

for $m = 1, \ldots, \frac{N+1}{2}$. Weighting with $P_{2m-1}(\mu)$ instead of μ^{2m-1} gives the same set of conditions, since the odd Legendre polynomials up to degree N and the odd monomials up to degree N span the same space; the two differ by an invertible mixing of the rows and have the same null space. The coefficients follow from $P_n(-\mu) = (-1)^n P_n(\mu)$,

$$c_{mn} = (-1)^{n+1}\,\frac{2n+1}{2}\int_0^1 \mu^{2m-1} P_n(\mu)\,d\mu \tag{11}$$

whose integrand is a polynomial of degree at most $2N$, so an $(N+1)$-point Gauss–Legendre rule mapped to $[0,1]$ evaluates it exactly rather than approximately.

Counting: (7) is $N+1$ first-order equations and needs $N+1$ conditions; (9) supplies $\frac{N+1}{2}$ and (10) the other $\frac{N+1}{2}$. For $N = 1$, (10) reads $-\frac{1}{4}\phi_0(a/2) + \frac{1}{2}\phi_1(a/2) = 0$, i.e. $\phi_0(a/2) = 2\phi_1(a/2)$: the classical statement that the scalar flux at a vacuum surface is twice the net current, and the origin of the extrapolation length $2/3$ quoted in Question 3.

Method 1: box discretisation and the Bell & Glasstone k iteration

The scheme. Divide $[0, a/2]$ into J cells of width $\Delta x = a/(2J)$ with nodes $x_j = j\Delta x$. Integrating (8) over one cell and closing it with the midpoint rule — derivative by the difference across the cell, value by the average of the two faces —

$$\left.\frac{d\phi_n}{dx}\right|_{j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} - \phi_{n,j}}{\Delta x}, \qquad \phi_{n,j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} + \phi_{n,j}}{2} \tag{12}$$

gives one $(N+1)$-row block per cell,

$$\left(\frac{\tilde{\mathbf{A}}}{\Delta x} + \frac{\mathbf{I}}{2}\right)\mathbf{\Phi}_{j+1} + \left(\frac{\mathbf{I}}{2} - \frac{\tilde{\mathbf{A}}}{\Delta x}\right)\mathbf{\Phi}_j = q_{j+\frac{1}{2}}\,\mathbf{e}_0, \qquad j = 0, \ldots, J-1 \tag{13}$$

This is the Keller box scheme, and it is the diamond difference of (26) written in moments rather than in ordinates: the same closure, applied to $N+1$ coupled unknowns per node instead of one per ordinate. Two consequences of that coupling: the sweep of Question 3 is not available, because $\tilde{\mathbf{A}}$ mixes moments that stream in both directions, so the cells must be solved simultaneously; and there is no negative-flux fixup, because ϕ_n for $n \geq 1$ is legitimately of either sign and a sign change carries no information.

The linear system. Stacking the $J+1$ nodal vectors gives $(J+1)(N+1)$ unknowns. Equation (13) supplies $J(N+1)$ rows, (9) supplies $\frac{N+1}{2}$ rows acting on $\mathbf{\Phi}_0$ and (10) the last $\frac{N+1}{2}$ acting on $\mathbf{\Phi}_J$; the count closes exactly. Ordered node by node the matrix is block-bidiagonal with two boundary blocks, i.e. banded with half-bandwidth $N+1$, and one banded (or sparse) LU factorisation solves it. Call that matrix $\mathbf{L}$: it carries streaming, collision and both boundary conditions, and it does *not* depend on k.

Why the source must stay on the right. The c/k term cannot be absorbed into $\mathbf{L}$. If it were, the system would be homogeneous and would admit a non-trivial solution only at the critical k — which is the answer, not a way of computing it. Left on the right-hand side it becomes a fission source lagged one iteration,

$$\mathbf{L}\,\mathbf{\Phi}^{(\ell+1)} = \frac{c}{k^{(\ell)}}\,\mathbf{F}\,\mathbf{\Phi}^{(\ell)}, \qquad q^{(\ell)}_{j+\frac{1}{2}} = \frac{c}{k^{(\ell)}} \cdot \frac{\phi^{(\ell)}_{0,j+1} + \phi^{(\ell)}_{0,j}}{2} \tag{14}$$

which is the power method on $\mathbf{L}^{-1}\mathbf{F}$, whose dominant eigenvalue is k_{eff}. Because c lumps scattering and fission into one isotropic term, there is no separate scattering source and hence no inner iteration at all: one factorisation, then one back-substitution per outer step, against the ten sweeps per outer that Question 5 needs.

The k update. Bell & Glasstone update k by the ratio of successive fission productions,

$$k^{(\ell+1)} = k^{(\ell)} \frac{\displaystyle\int_0^{a/2} \phi_0^{(\ell+1)}\,dx}{\displaystyle\int_0^{a/2} \phi_0^{(\ell)}\,dx} \tag{15}$$

the constant c cancelling between numerator and denominator. The integral is taken as $\Delta x \sum_j \phi_{0,j+1/2}$, which is the trapezoid rule on the nodes; this matters, because it is the same midpoint average the source (13) uses, so (15) is exactly the Rayleigh ratio of the discrete operator and not a second, inconsistent discretisation of the same integral. The iterate is then rescaled by $\max_j \phi_{0,j}$, which has no effect on (15) — it only keeps the vector from drifting towards 0 or ∞ as k^ℓ compounds — and the loop stops on $|k^{(\ell+1)} - k^{(\ell)}| < \epsilon_k$. Convergence is linear at the dominance ratio of the two leading eigenvalues, which for a bare slab is well separated, and the starting guess

$$\phi_0^{(0)}(x) = \cos\left(\frac{\pi x}{2\,(a/2 + 0.7104)}\right), \qquad k^{(0)} = 1 \tag{16}$$

is the diffusion fundamental mode with the asymptotic extrapolation distance, already almost the answer.

The outer loop. Dividing the fission source by k makes a solution exist at every thickness, so criticality is a root search on

$$f(a/2) = k(a/2) - 1 \tag{17}$$

exactly as in Questions 3–5. $k(a/2)$ is monotone increasing — a thicker slab leaks a smaller fraction — and runs from $k \to 0$ as $a/2 \to 0$ to the infinite-medium value $k \to c$ as $a/2 \to \infty$. So for every $c > 1$ a root exists and is unique, and any bracket $[(a/2)_{\text{lo}}, (a/2)_{\text{hi}}]$ straddling the diffusion estimate below is admissible; `brentq` closes it in a handful of evaluations, each of which is one full power iteration.

Errors. The box scheme is second order, so the discretisation error in $(a/2)_{\text{crit}}$ falls as Δx^2 and is separable from the truncation error in N by running the same N on a refined mesh — the check reported in Question 3 for S_N, and the quantity that Method 2 removes altogether.

Method 2: modal elimination, as an N-exact benchmark

Method 1 carries two errors, one in N and one in Δx. Method 2 removes the second by solving (8) in closed form at $k = 1$ and imposing criticality directly on the boundary conditions.

Split the $N+1$ moments into $M = \frac{N+1}{2}$ even and M odd, $\mathbf{\Phi}_{\text{even}} = (\phi_0, \phi_2, \dots)$ and $\mathbf{\Phi}_{\text{odd}} = (\phi_1, \phi_3, \dots)$. Every derivative in (7) changes the parity of its index, so the even-n and odd-n equations separate:

$$\mathbf{A}\frac{d\mathbf{\Phi}_{\text{odd}}}{dx} + \mathbf{\Sigma}_0(k)\,\mathbf{\Phi}_{\text{even}} = 0, \qquad \mathbf{B}\frac{d\mathbf{\Phi}_{\text{even}}}{dx} + \mathbf{\Phi}_{\text{odd}} = 0, \qquad \mathbf{\Sigma}_0(k) = \text{diag}\big(1 - \tfrac{c}{k}, 1, \dots, 1\big) \tag{18}$$

$\mathbf{A}$ and $\mathbf{B}$ being the two off-diagonal blocks of $\tilde{\mathbf{A}}$ under that reordering, $\tilde{\mathbf{A}} = \left(\begin{smallmatrix} 0 & \mathbf{A} \\ \mathbf{B} & 0 \end{smallmatrix}\right)$, with rows $A_{m,\cdot}$ from $n = 2m$ and $B_{m,\cdot}$ from $n = 2m+1$. Substituting $\boldsymbol{\Phi}_{\text{odd}} = -\mathbf{B}\,\boldsymbol{\Phi}'_{\text{even}}$ into the first equation eliminates the odd moments and leaves a coupled Helmholtz system,

$$\frac{d^2\boldsymbol{\Phi}_{\text{even}}}{dx^2} + \mathbf{K}^2(k)\,\boldsymbol{\Phi}_{\text{even}} = 0, \qquad \mathbf{K}^2(k) = -\,(\mathbf{AB})^{-1}\boldsymbol{\Sigma}_0(k) \tag{19}$$

The minus sign is not cosmetic and is the easiest thing to lose: the elimination produces $\mathbf{AB}\,\boldsymbol{\Phi}''_{\text{even}} = \boldsymbol{\Sigma}_0\boldsymbol{\Phi}_{\text{even}}$, and moving that to the form (19) flips it. At $N = 1$ it reads $\phi_0'' + 3(c-1)\phi_0 = 0$, the classic diffusion buckling $B = \sqrt{3(c-1)}$ of the table in Question 1 — with the sign dropped one gets cosh where cos belongs and no critical thickness at all.

The spectrum of $\mathbf{K}^2$. Since $\tilde{\mathbf{A}}^2 = \left(\begin{smallmatrix} \mathbf{AB} & 0 \\ 0 & \mathbf{BA} \end{smallmatrix}\right)$ and the eigenvalues of $\tilde{\mathbf{A}}$ are the $N+1$ roots $\pm\mu_i$ of P_{N+1}, the matrix $\mathbf{AB}$ has the M eigenvalues $\mu_i^2 > 0$ and is similar to a positive definite matrix. By Sylvester's law of inertia the eigenvalues of (19) then carry the signs of $-\boldsymbol{\Sigma}_0$, which for $c > 1$ at $k = 1$ has one negative and $M-1$ positive entries: **exactly one eigenvalue $\lambda_1^2 = B_0^2 > 0$, and $M-1$ eigenvalues $\lambda_j^2 = -\kappa_j^2 < 0$.** The one oscillatory mode is the fundamental; the rest are the boundary layers, and at $c = 1$ the positive one collapses to $B_0 = 0$, an infinite critical slab.

Assembling the criticality condition. Diagonalise $\mathbf{K}^2(1) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$. The solution even about $x = 0$, which satisfies (9) identically because $\boldsymbol{\Phi}_{\text{odd}} = -\mathbf{B}\boldsymbol{\Phi}'_{\text{even}}$ vanishes wherever $\boldsymbol{\Phi}'_{\text{even}}$ does, is

$$\boldsymbol{\Phi}_{\text{even}}(x) = C_1\mathbf{v}_1\cos(B_0x) + \sum_{j=2}^{M} C_j\mathbf{v}_j\cosh(\kappa_jx) \tag{20}$$

$$\boldsymbol{\Phi}_{\text{odd}}(x) = C_1(\mathbf{B}\mathbf{v}_1)B_0\sin(B_0x) - \sum_{j=2}^{M} C_j(\mathbf{B}\mathbf{v}_j)\kappa_j\sinh(\kappa_jx) \tag{21}$$

M free constants against the M Marshak conditions (10). Splitting the coefficient matrix c_{mn} into its even and odd columns $\mathbf{M}_{\text{even}}, \mathbf{M}_{\text{odd}}$ and evaluating at $x = a/2$ gives M homogeneous equations $\mathbf{H}(a/2)\mathbf{C} = \mathbf{0}$ with

$$\begin{aligned}
\mathbf{H}_{\cdot 1} &= \mathbf{M}_{\text{even}}\mathbf{v}_1\cos\!\left(\tfrac{B_0a}{2}\right) + \mathbf{M}_{\text{odd}}(\mathbf{B}\mathbf{v}_1)\,B_0\sin\!\left(\tfrac{B_0a}{2}\right) \\
\mathbf{H}_{\cdot j} &= \mathbf{M}_{\text{even}}\mathbf{v}_j\cosh\!\left(\tfrac{\kappa_ja}{2}\right) - \mathbf{M}_{\text{odd}}(\mathbf{B}\mathbf{v}_j)\,\kappa_j\sinh\!\left(\tfrac{\kappa_ja}{2}\right), \qquad j \geq 2
\end{aligned}$$

and a non-trivial flux exists only where

$$\boxed{\det\mathbf{H}(a/2) = 0} \tag{22}$$

$(a/2)_{\text{crit}}$ is its smallest positive root. Two numerical points: the cosh and sinh columns overflow once $\frac{1}{2}\kappa_ja \gtrsim 700$, so each is divided by its own $\cosh(\frac{1}{2}\kappa_ja)$, which leaves $\mathbf{v}_j$ and $-(\mathbf{B}\mathbf{v}_j)\kappa_j\tanh(\frac{1}{2}\kappa_ja)$ and cannot move a zero of the determinant, a column scaling being a non-zero factor of it; and the determinant itself spans many orders of magnitude across the bracket, so it is the sign change that is tracked, not the value.

The two methods against P_1 in closed form

At $N = 1$ everything above is elementary and both methods must reproduce it. With $B = \sqrt{3(c-1)}$ and $\phi_0 = \cos(Bx)$, the Marshak condition $\phi_0(a/2) = 2\phi_1(a/2)$ and $\phi_1 = -\frac{1}{3}\phi_0'$ give $\cot(Ba/2) = \frac{2}{3}B$, that is

$$\left(\frac{a}{2}\right)_{\text{crit}}^{P_1} = \frac{1}{B}\arctan\!\left(\frac{3}{2B}\right) \tag{23}$$

c	$B = \sqrt{3(c-1)}$	P_1 Marshak	P_1 Mark	S_2 of Q3	exact transport
1.2	0.774597	1.412499	1.484982	1.48499	1.28981
1.5	1.224745	0.723479	0.780013	0.78001	0.60762
1.8	1.549193	0.496559	0.542907	0.54291	0.39314

Table 7: Analytic P_1 half-thickness in mean free paths, against Question 3's S_2 column. The Mark column, $\arctan(\sqrt{3}/B)/B$, is the same equation with the extrapolation length $1/\sqrt{3}$ in place of $2/3$, and it reproduces the S_2 entry to five digits — the demonstration promised in Question 3 that P_N and S_{N+1} differ in slab geometry by their boundary condition alone. Both are the $N = 1$ target that Method 1 must approach as $\Delta x \to 0$ and that Method 2 must return outright.

Results

c	order	Method 1 [mfp]	Method 2 [mfp]	gap	exact transport [mfp]	departure
1.2	P_1	1.412502	1.412499	$2.5 \cdot 10^{-6}$	1.28981	+9.51%
1.2	P_3	1.302001	1.301998	$2.5 \cdot 10^{-6}$	1.28981	+0.94%
1.2	P_5	1.292597	1.292594	$2.4 \cdot 10^{-6}$	1.28981	+0.22%
1.2	P_9	1.290382	1.290379	$2.4 \cdot 10^{-6}$	1.28981	+0.04%
1.5	P_1	0.723480	0.723479	$1.6 \cdot 10^{-6}$	0.60762	+19.07%
1.5	P_3	0.629149	0.629148	$1.7 \cdot 10^{-6}$	0.60762	+3.54%
1.5	P_5	0.612136	0.612135	$1.7 \cdot 10^{-6}$	0.60762	+0.74%
1.5	P_9	0.606467	0.606466	$1.7 \cdot 10^{-6}$	0.60762	−0.19%
1.8	P_1	0.496560	0.496559	$1.2 \cdot 10^{-6}$	0.39314	+26.30%
1.8	P_3	0.418210	0.418209	$1.3 \cdot 10^{-6}$	0.39314	+6.38%
1.8	P_5	0.399736	0.399735	$1.4 \cdot 10^{-6}$	0.39314	+1.68%
1.8	P_9	0.391130	0.391129	$1.3 \cdot 10^{-6}$	0.39314	−0.51%

Table 8: Critical half-thickness $(a/2)_{\text{crit}}$ in mean free paths. Method 1 is the box scheme on 200 cells; Method 2 is $\det \mathbf{H}(a/2) = 0$. The gap between them is the relative difference, and the departure is Method 2's against the exact-transport reference $a/2 = \frac{\pi}{2}\nu_0 - z_0$ of Assignment 1.

The two methods against each other. The gap column is the only free check available, since the two share nothing but the matrices of (8) and (10): one iterates a discretised operator, the other never discretises space at all. They agree to between one and three parts in 10^6, and that residue is Method 1's Δx^2 and nothing else. Refining the mesh at $c = 1.5$, $N = 5$ against Method 2's 0.612134509:

cells J	25	50	100	200	400
error [mfp]	$6.61 \cdot 10^{-5}$	$1.65 \cdot 10^{-5}$	$4.13 \cdot 10^{-6}$	$1.03 \cdot 10^{-6}$	$2.58 \cdot 10^{-7}$
ratio	—	4.00	4.00	4.00	4.00

Each halving of Δx divides the error by 4.00 to three digits, which is the second order claimed above measured rather than asserted, and it confirms that the boundary rows are as accurate as the interior ones — a first-order boundary closure would have shown up here as a ratio of 2.

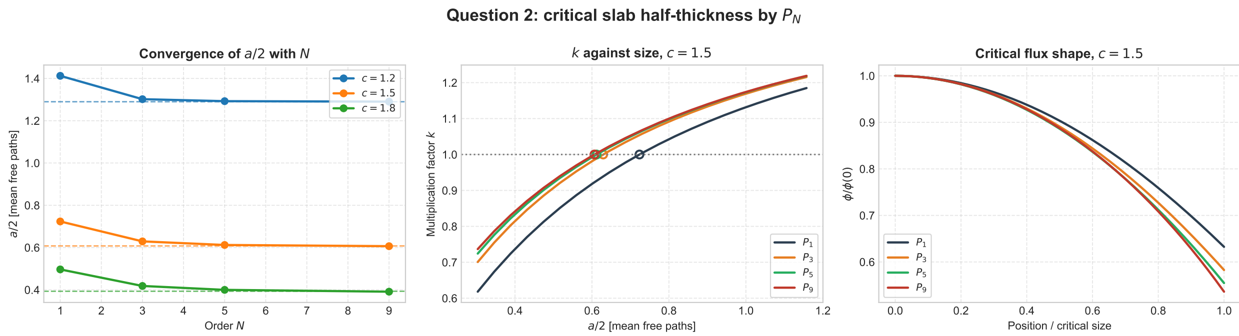

Figure 3: Left: convergence of $(a/2)_{\text{crit}}$ with N, dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings, which for P_3, P_5 and P_9 are already almost on top of one another. Right: the normalised critical flux — as in Question 3, the order changes the surface value and not the interior shape.

Where the reference column is wrong. The last two departures are negative, and a P_N half-thickness below the true one would be a real defect. It is the reference that is at fault: $a/2 = \frac{\pi}{2}\nu_0 - z_0$ is the asymptotic relation, accurate to about 1.5% in z_0 as Question 4 measures. Against the exact one-speed value at $c = 1.5$, the Sood *et al.* 0.605055 mfp used in Question 3, the P_N sequence is monotone from *above* and does not cross:

P_1	P_3	P_5	P_9	P_{11}	P_{15}	P_{31}
0.723479	0.629148	0.612135	0.606466	0.605952	0.605550	0.605181

P_9 is +0.23% above it and P_{31} is +0.02%. Every entry of the table is therefore converging correctly; the sign of the last column is an artefact of comparing against a reference that is itself 0.4% too large at this c.

The S_N method of Questions 3–5

All three questions solve the same monoenergetic, steady-state problem,

$$\mu\frac{\partial\psi}{\partial x} + \Sigma_t\psi = \frac{1}{2}\left[\Sigma_s\phi + \frac{1}{k}\nu\Sigma_f\phi\right], \qquad \phi(x) = \int_{-1}^{1}\psi(x,\mu)\,d\mu \tag{24}$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^{N} w_m\psi_{i,m} \tag{25}$$

Space, slab. Integrating over cell i and closing with the diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i+1/2,m} + \psi_{i-1/2,m})$ gives the sweep

$$\psi_{i,m} = \frac{2|\mu_m|\,\psi_{\text{in},m} + \frac{1}{2}S_i\Delta x}{\Sigma_t\Delta x + 2|\mu_m|}, \qquad \psi_{\text{out},m} = 2\psi_{i,m} - \psi_{\text{in},m} \tag{26}$$

swept from $x = a/2$ inwards for $\mu_m < 0$ and from $x = 0$ outwards for $\mu_m > 0$.

Space, sphere. The spherical equation carries an angular redistribution term,

$$\mu \frac{\partial \psi}{\partial r} + \frac{1-\mu^2}{r}\frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \tag{27}$$

whose conservative form integrates over the cell to

$$\mu_m \left(A_{i+\frac{1}{2}}\psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}}\psi_{i-\frac{1}{2},m}\right) + \frac{A_{i+\frac{1}{2}} - A_{i-\frac{1}{2}}}{w_m}\left(\alpha_{m+\frac{1}{2}}\psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}}\psi_{i,m-\frac{1}{2}}\right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2}V_i \tag{28}$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r^3_{i+1/2} - r^3_{i-1/2})$. This is (26) with a second outgoing face, in angle, closed by the same diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i,m+1/2} + \psi_{i,m-1/2})$. The angular coefficients follow from demanding that a flat flux solve (28) exactly:

$$\alpha_{\frac{1}{2}} = 0, \qquad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m\mu_m, \qquad \alpha_{N+\frac{1}{2}} = 0 \ \text{ since } \sum_m w_m \mu_m = 0 \tag{29}$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1-\mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is the plain slab sweep at $|\mu| = 1$, run inwards from the vacuum boundary.

One cell solve for both geometries.

Equations (26) and (28) are the same equation over a different number of faces, and it is worth writing that form down once because it is what the code implements. Let a cell carry a list of outgoing faces f, each contributing $a^f_{\text{out}}\psi^f_{\text{out}} - a^f_{\text{in}}\psi^f_{\text{in}}$ to the balance:

$$\sum_f \left(a^f_{\text{out}}\psi^f_{\text{out}} - a^f_{\text{in}}\psi^f_{\text{in}}\right) + \Sigma_t V \psi = \tfrac{1}{2} S\, V \tag{30}$$

Closing *every* face with the same diamond relation $\psi^f_{\text{out}} = 2\psi - \psi^f_{\text{in}}$ and collecting ψ gives one formula for any geometry:

$$\psi = \frac{\frac{1}{2}SV + \sum_f \left(a^f_{\text{out}} + a^f_{\text{in}}\right)\psi^f_{\text{in}}}{\Sigma_t V + 2\sum_f a^f_{\text{out}}} \tag{31}$$

The two geometries differ only in what they put in the list:

	faces	a_{out}	a_{in}
slab	space	$\lvert\mu_m\rvert$	$\lvert\mu_m\rvert$
sphere	space	$\lvert\mu_m\rvert A_{\text{out}}$	$\lvert\mu_m\rvert A_{\text{in}}$
	angle	$\frac{A_{i+1/2}-A_{i-1/2}}{w_m}\,\alpha_{m+\frac{1}{2}}$	$\frac{A_{i+1/2}-A_{i-1/2}}{w_m}\,\alpha_{m-\frac{1}{2}}$

with $V = \Delta x$ in the slab. Substituting the slab row into (31) returns (26) exactly, and the sphere's two rows return (28).

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{\text{in},m}$ comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. This never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t \Delta x/|\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu\Sigma_{f,i}\phi_i^{(l)}/k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i}\phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)}\,\frac{P^{(l+1)}}{P^{(l)}}, \qquad P^{(l)} = \sum_i \nu\Sigma_{f,i}\,\phi_i^{(l)}\,V_i \tag{32}$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k.

The meaning of c. Questions 3 and 4 specify only $c = (\Sigma_s + \nu\Sigma_f)/\Sigma_t$. At $k = 1$ the source bracket collapses to $c\,\Sigma_t\phi$ whatever the split between scattering and fission, so the critical size depends on c alone and the code takes the simplest split, $\Sigma_t = 1$, $\Sigma_s = 0$, $\nu\Sigma_f = c$. Question 5 has no such freedom and uses the tabulated cross sections directly.

Question 3: Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2}\nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0.

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 9: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640$ cm^{-1}, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10}, 0.606407 at S_{16}, 0.605631 at S_{24}, 0.605195 at S_{48}.

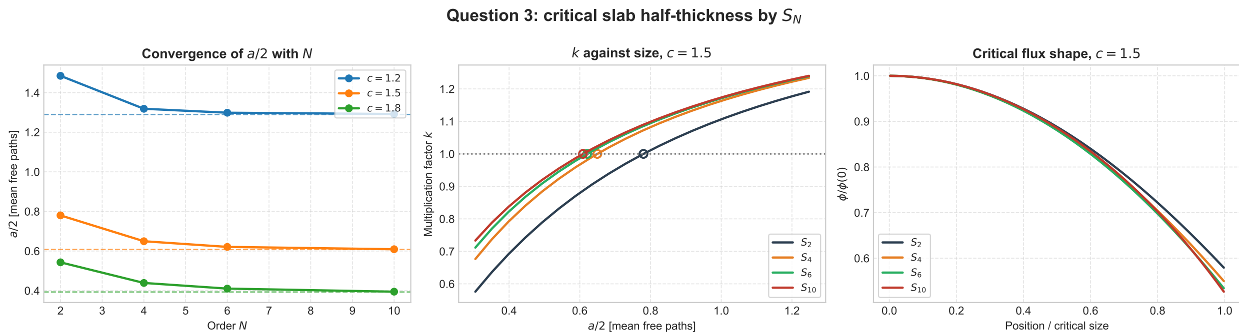

Figure 4: Left: convergence of $a/2$ with N, dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Right: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N + 1$ integrates polynomials up to degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*'s, $\int_0^1 P_{2j+1}(\mu)\psi\, d\mu = 0$, weight the whole incoming half-range; Mark's pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree; with Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

Question 2 supplies the rest of that comparison, since it solves the same slab at $N = 1, 3, 5, 9$ under Marshak conditions:

Three things in that table follow from the argument above. **The sign of the gap is the sign of l_0**: Marshak's extrapolation 2/3 is longer than Mark's $1/\sqrt{3} = 0.577$, so the P_N flux is driven to zero further outside the slab and less material is needed — P_N is below S_{N+1} at every one of the twelve entries. **The gap is a boundary effect, so it shrinks with N**, from 5–9% at $N = 1$ to under 1% at $N = 9$; by then both boundary conditions are resolving the same half-range flux and the choice between them stops mattering. And **it grows with c** at fixed N — 5.13%, 7.81%, 9.33% at $N = 1$ — because a larger c means a thinner slab, a shorter relaxation length and a more forward-peaked surface flux, which is exactly the shape a half-range condition of low order cannot capture. The whole difference is worth at most a factor of two in either method's error: at $c = 1.5$, P_9 is +0.23% from the true 0.605055 mfp where S_{10} is +0.66%, so a P_N solve buys roughly the accuracy of an S_N solve of the next order up.

A second source of difference is that S_N is a collocation method: it satisfies the equation exactly along N directions and says nothing between them. At the vacuum face the leaked current $\sum_{\mu_m>0} w_m \mu_m \psi_m$ is that of a flux strongly peaked towards $\mu = 1$, which a coarse quadrature underestimates — so the slab has to be made *thicker* than it should be to stay critical, and the table comes down from above.

c	pair	P_N (Marshak)	S_{N+1} (Mark)	$S_{N+1}/P_N - 1$	exact transport
1.2	P_1 / S_2	1.41250	1.48499	+5.13%	1.28981
1.2	P_3 / S_4	1.30200	1.31855	+1.27%	1.28981
1.2	P_5 / S_6	1.29259	1.29861	+0.47%	1.28981
1.2	P_9 / S_{10}	1.29038	1.29218	+0.14%	1.28981
1.5	P_1 / S_2	0.72348	0.78001	+7.81%	0.60762
1.5	P_3 / S_4	0.62915	0.64949	+3.23%	0.60762
1.5	P_5 / S_6	0.61213	0.62089	+1.43%	0.60762
1.5	P_9 / S_{10}	0.60647	0.60904	+0.42%	0.60762
1.8	P_1 / S_2	0.49656	0.54291	+9.33%	0.39314
1.8	P_3 / S_4	0.41821	0.43875	+4.91%	0.39314
1.8	P_5 / S_6	0.39974	0.41014	+2.60%	0.39314
1.8	P_9 / S_{10}	0.39113	0.39459	+0.88%	0.39314

Table 10: The two methods at matched order, in mean free paths. Every P_N entry is below its S_{N+1} partner and closer to the exact value, and the gap closes as N grows.

All of that is a statement about *slabs*. In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r}\frac{\partial\psi}{\partial\mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (28) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions. At $c = 1.5$, S_2 gives $R_c = 1.607$ mean free paths where P_1 with the Mark extrapolation gives 1.919 and with Marshak's 1.822 — a spread no choice of boundary condition can close.

Question 4: Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (28). The reference is Assignment 1's $\Sigma_t R_c = \pi\nu_0 - z_0$.

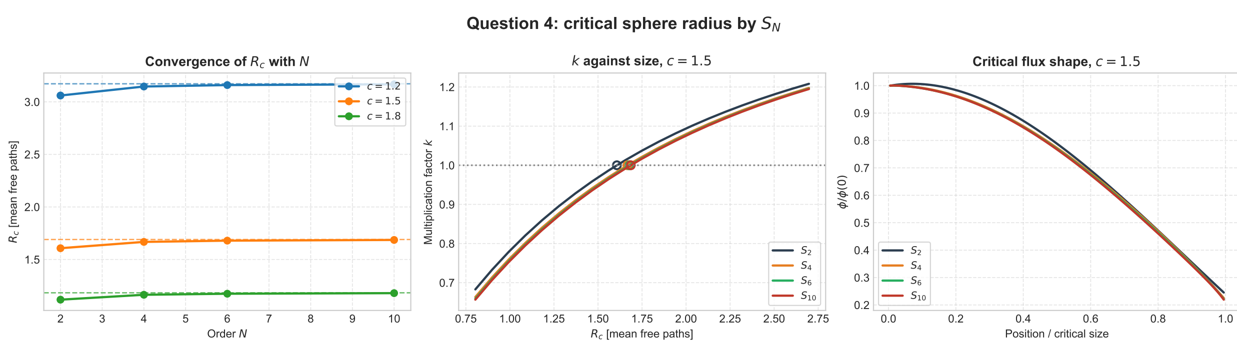

Figure 5: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	-3.51%
1.2	S_4	3.14631	3.17204	-0.81%
1.2	S_6	3.15991	3.17204	-0.38%
1.2	S_{10}	3.16728	3.17204	-0.15%
1.5	S_2	1.60725	1.69010	-4.90%
1.5	S_4	1.66739	1.69010	-1.34%
1.5	S_6	1.67934	1.69010	-0.64%
1.5	S_{10}	1.68595	1.69010	-0.25%
1.8	S_2	1.11753	1.18295	-5.53%
1.8	S_4	1.16413	1.18295	-1.59%
1.8	S_6	1.17412	1.18295	-0.75%
1.8	S_{10}	1.17971	1.18295	-0.27%

Table 11: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2}\nu_0$ and $R_c + z_0 = \pi\nu_0$, and therefore

$$z_0 = R_c - 2\,(a/2) \tag{33}$$

which lets the two tables above check each other without any further calculation:

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 12: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

Question 5: Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640$ cm^{-1} for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3\rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c-only problems above. The orders are S_2 to S_{16} rather than the S_2–S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	−5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	−1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	−0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	−0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	−4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	−1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	−0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	−0.15%

Table 13: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

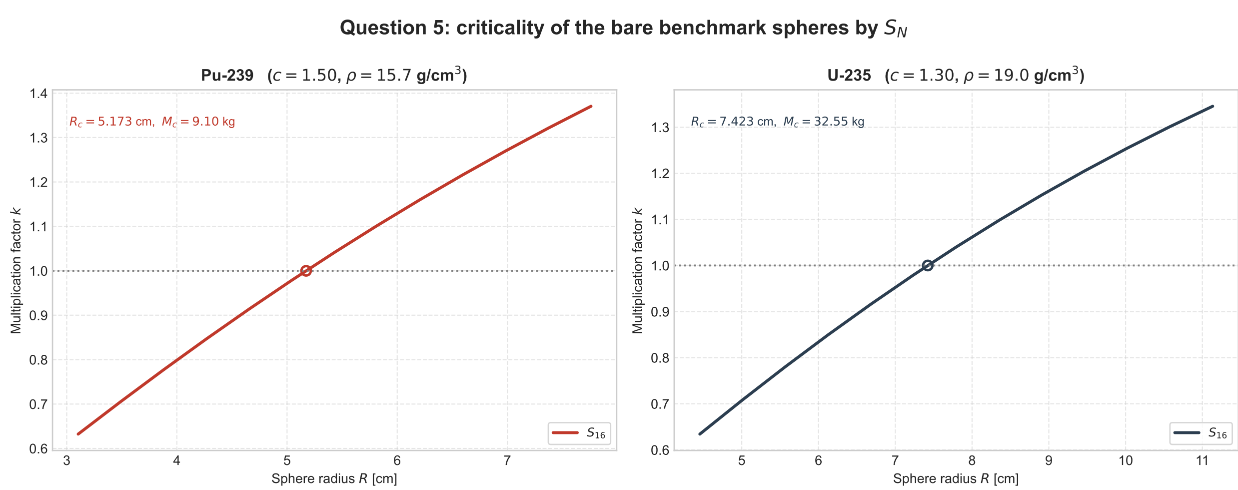

Figure 6: k against sphere radius at S_{16}, with the $k = 1$ crossing that fixes the critical mass.

approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.